Analysis of Presidential Lifespans – A Report

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Findings

Creating and running this analysis program produced many interesting results. Looking at the 10 shortest lifespans, many involved presidents who died in office (or shortly after, in the case of James K. Polk) Also, the four presidents who were assassinated in office (Lincoln, Garfield, McKinley, and Kennedy) all appeared in the 10 shortest lifespans. Of the 10 longest lifespans, 6 belonged to presidents who served in the 20th century or later, reflecting medical advancements in treating diseases associated with age. The data given is out of date, as it does not reflect Jimmy Carter's death, but he would be on the top of the list of longest lifespans, having lived to 100.

Tables and Assumptions

When calculating lifespans, I only included presidents who have death dates listed, and therefore whose lifespans are currently known. I used Python for the analysis portion, as it is a language I have much more experience with as opposed to R.

I first created these tables showing the presidents with the 10 longest and 10 shortest lifespans (excluding presidents listed as still alive) and displayed them, alongside the exact lifespan in years, months, and days (using the relativedelta module for improved date math) (Figure 1):

10 Longest Lifespans

PRESIDENT	years_lived	months_lived	days_lived
George Bush	94	5	18
Gerald Ford	93	5	12
Ronald Reagan	93	3	30
John Adams	90	8	4
Herbert Hoover	90	2	10
Harry S. Truman	88	7	18
James Madison	85	3	12
Thomas Jefferson	83	2	21
Richard Nixon	81	3	13
John Quincy Adams	80	7	12

10 Shortest Lifespans

PRESIDENT	years_lived	months_lived	days_lived
John F. Kennedy	46	5	24
James A. Garfield	49	10	0
James K. Polk	53	7	13
Abraham Lincoln	56	2	3
Chester Arthur	57	1	13
Warren G. Harding	57	9	0
William McKinley	58	7	16
Theodore Roosevelt	60	2	10
Calvin Coolidge	60	6	1
Franklin Roosevelt	63	2	13

Figure 1: Tables with 10 longest and 10 shortest lifespans

I then used numpy to calculate the mean, median, minimum, maximum, and standard deviation of the lifespan data in days. For the mode, I used scipy's stats module, and only took into account the age in years of the president at death, as no two presidents lived the exact same number of days (so the mode function on the lifespans in days simply gave the lowest value, which was not useful). I then created another table with these statistical values (Figure 2):

Statistics

Calculation	Value
Mean	~71 year(s), 5 month(s), 2 day(s)
Median	~70 year(s), 3 month(s), 3 day(s)
Mode	67 years
Max	~94 year(s), 10 month(s), 14 day(s)
Min	~46 year(s), 1 month(s), 8 day(s)
StdDev	~12 year(s), 7 month(s), 7 day(s)

Figure 2: Some values produced by the analysis

Finally, for the table, I used the matplotlib library as I have worked with it in the past. I created this table, with horizontal bars showing the lifespan in years, going from shortest on the bottom to longest at the top. I also added vertical lines showing the mean and median, so that the lifespan of any president can be easily compared to it. I used a lighter color for the bars and a darker color for the mean and median lines to improve readability (Figure 3).

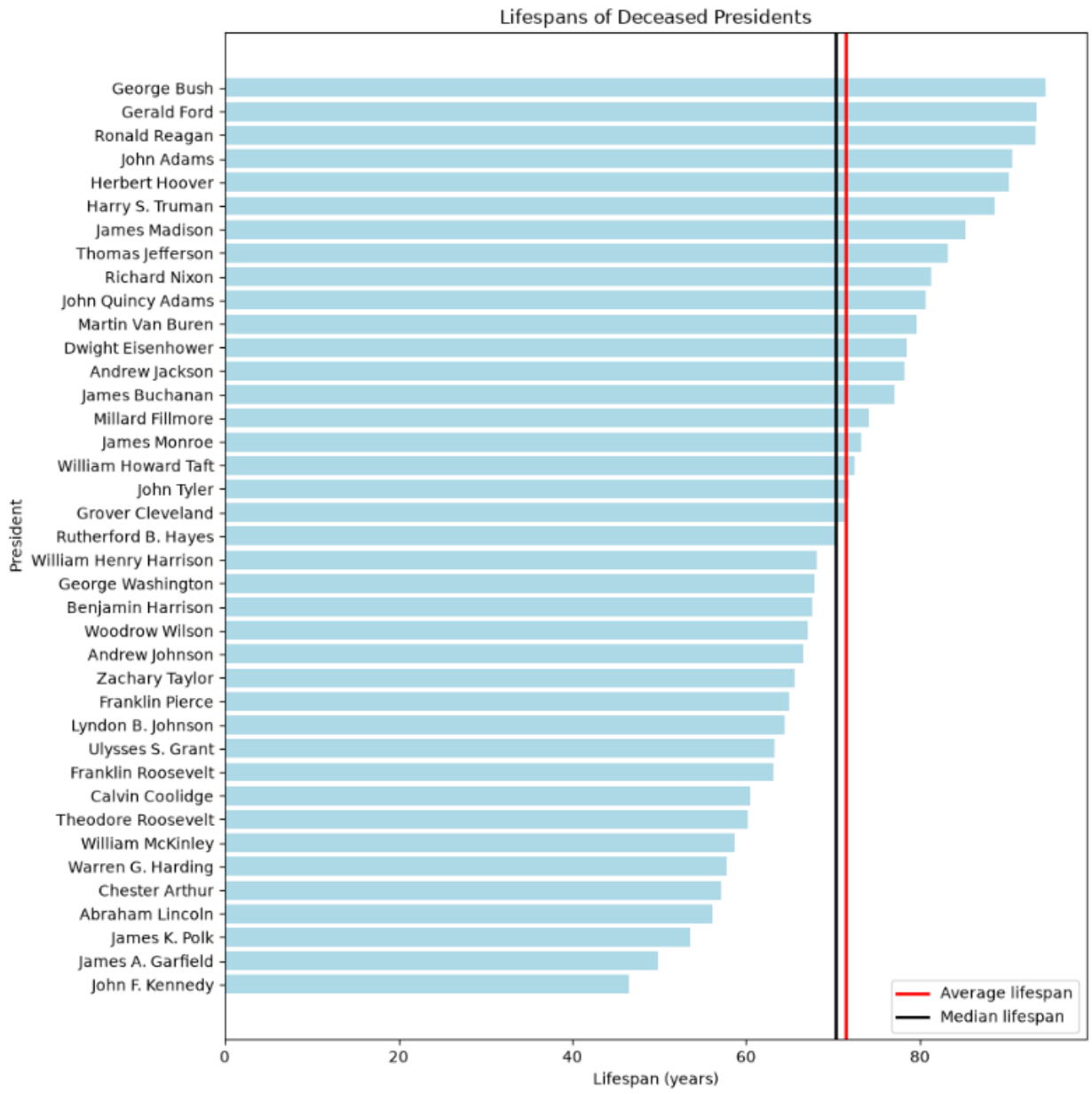


Figure 3: Visual table of lifespans of deceased presidents

Conclusion

Looking at the table reveals a relatively even distribution of lifespans. George Bush had the longest lifespan listed in this data, at 94 years, 5 months, and 18 days. The shortest lifespan belonged to John F. Kennedy, at just 46 years, 5 months, and 24 days. This means there was a range of just below 48 years in lifespans. Of the presidents who were not assassinated, the president with the shortest lifespan was James K. Polk, whose lifespan was 53 years, 7 months, and 13 days.